

11.5 Alternating Series



Definition

- **An alternating series is a series whose terms are alternately positive and negative.**

(2)

- **Example:**

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$

$$-\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7} - \cdots = \sum_{n=1}^{\infty} (-1)^{n} \frac{n}{n+1}$$



Alternating Series

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- In this section we learn how to deal with series whose terms are not necessarily positive. Of particular importance are *alternating series,* whose terms alternate in sign.

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n}$$

- An **alternating series** is a series whose terms are alternately positive and negative. Here are two

$$-\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \frac{5}{6} + \frac{6}{7} - \cdots = \sum_{n=1}^{\infty} (-1)^{n} \frac{n}{n+1}$$



Ex: Determine whether the following series are alternating or not. (4)

1. $\sum_{n=1}^{\infty} (-1)^{2n} \frac{n}{n^2+1}$ ← this example is not an alternating series

2. $\sum_{n=1}^{\infty} \frac{\cos n\pi}{n} = -1 + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \cdots$

So, it is an alternating series.



Alternating Series

- We see from these examples that the *n*th term of an alternating series is of the form

- $a_n = (-1)^{n-1} b_n$ or $a_n = (-1)^n b_n$

- where b_n is a positive number. (In fact, $b_n = |a_n|$.)

- The following test says that if the terms of an alternating series decrease toward 0 in absolute value, then the series converges.



Remark: The series

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$$ is convergent series

while

$$\sum_{n=1}^{\infty} \frac{1}{n}$$ (is Harmonic series)

divergent



Alternating Series

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Alternating Series Test If the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n = b_1 - b_2 + b_3 - b_4 + b_5 - b_6 + \cdots \qquad b_n > 0$$

satisfies

(i) $b_{n+1} \leqslant b_n$ for all n

(ii) $\lim_{n \to \infty} b_n = 0$

then the series is convergent.

Example:

$$\sum_{n=2}^{\infty} (-1)^n \frac{5n+3}{2n-7}$$

ignore $(-1)^n$ when you test the series

(i) $b_{n+1} \leq b_n$ for all n

(ii) $\lim_{n\to\infty} b_n = 0$

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Sol: $\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{5n+3}{2n-7} = \frac{\infty}{\infty}$ ind. Form

$$= \lim_{n\to\infty} \frac{\frac{5n}{n} + \frac{3}{n}}{\frac{2n}{n} - \frac{7}{n}} = \lim_{n\to\infty} \frac{5+\frac{3}{n}}{2-\frac{7}{n}} = \frac{5}{2} \neq 0$$

So this condition is fail to satisfy $\Rightarrow$ it is div.



Example

- **Determine whether the following series convergent or divergent**
  - $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$
  - $\sum_{n=1}^{\infty} \frac{(-1)^n 3n}{4n - 1}$



Ex: $\sum (-1)^n \frac{n}{\ln n}$

$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n}{\ln n} = \frac{\infty}{\infty}$ ind. Form, so we will use L'H. Rule

(i) $b_{n+1} \leq b_n$ for all n

(ii) $\lim_{n\to\infty} b_n = 0$

$= \lim_{n\to\infty} \frac{n'}{(\ln n)'} = \lim_{n\to\infty} \frac{1}{\frac{1}{n}} =$

$= \lim_{n\to\infty} n = \infty \neq 0$

$\Rightarrow$ this series is div.

Ex: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$

Sol:

1. $\lim_{n\to\infty} \frac{1}{n!} = \frac{1}{\infty!} = \frac{1}{\infty} = 0$ ✓

2. $b_{n+1} \leq b_n$ (the series is dec.)

$$\frac{1}{(n+1)!} \leq \frac{1}{n!}$$ ✓

$$\frac{1}{5!} \leq \frac{1}{4!}$$

$$\frac{1}{120} \leq \frac{1}{24}$$

$n! = n.(n-1).(n-2).\cdots.1$

$4! = 4\cdot 3\cdot 2\cdot 1 = 24$

$5! = 5.4.3.2.1 = 120$

$(n+1)! = (n+1)(n)(n-1)(n-2)\cdots 1$

$\Rightarrow (n+1)! = (n+1)\,n!$

as $\frac{1}{(n+1).n!} \leq \frac{1}{n!}$

So, this condition is hold.

$\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ is conv.

Example:

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Test the series $\sum_{n=1}^{\infty} (-1)^{n+1} \dfrac{n^2}{n^3+1}$ for convergence or divergence.

$n \in [1, \infty)$

Sol:

1. $\lim_{n\to\infty} \dfrac{n^2}{n^3+1} = \dfrac{\infty}{\infty}$ ind. Form

$\lim_{n\to\infty} \frac{1}{n} = 0$

$$= \lim_{n\to\infty} \frac{\frac{n^2}{n^3}}{\frac{n^3}{n^3} + \frac{1}{n^3}} = \lim_{n\to\infty} \frac{\frac{1}{n}}{1 + \frac{1}{n^3}}$$

$$= \frac{0}{1+0} = 0 \quad \checkmark$$

(2) $b_{n+1} \leq b_n$ [instead of that we can just prove that the given series is div]

$$f(n) = \frac{n^2}{n^3+1} = \frac{u}{v}$$

$$f'(n) = \frac{(n^3+1)(2n) - (3n^2)\, n^2}{(n^3+1)^2}$$

$$= \frac{2n^4 + 2n - 3n^4}{(n^3+1)^2}$$

$$= \frac{-n^4 + 2n}{(n^3+1)^2} = \frac{n(-n^3+2)}{(n^3+1)^2}$$

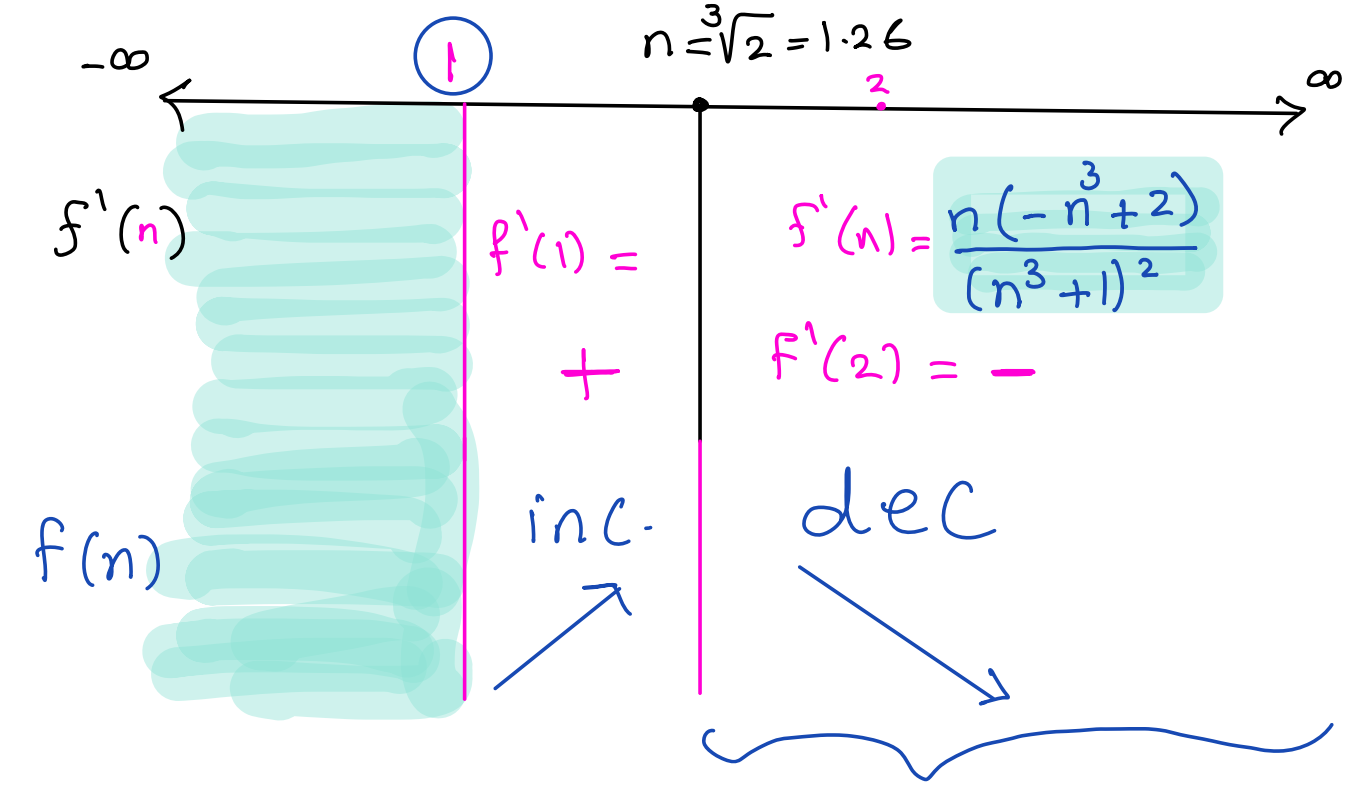
$= 0 \iff$ the num. $= 0$

$\Leftrightarrow n.(-n^3+2)=0$

$\Leftrightarrow$ either $n=0$ (as $n \in [1,\infty)$) or $(-n^3+2)=0$

$\Rightarrow -n^3+2=0 \Leftrightarrow n^3=2 \Leftrightarrow n=\sqrt[3]{2}$

$\Leftrightarrow n=1.26$



$\Rightarrow$ our series is dec on $[2,\infty)$ which mean the second condition is hold $\Rightarrow$ our series is conv.



Example 1

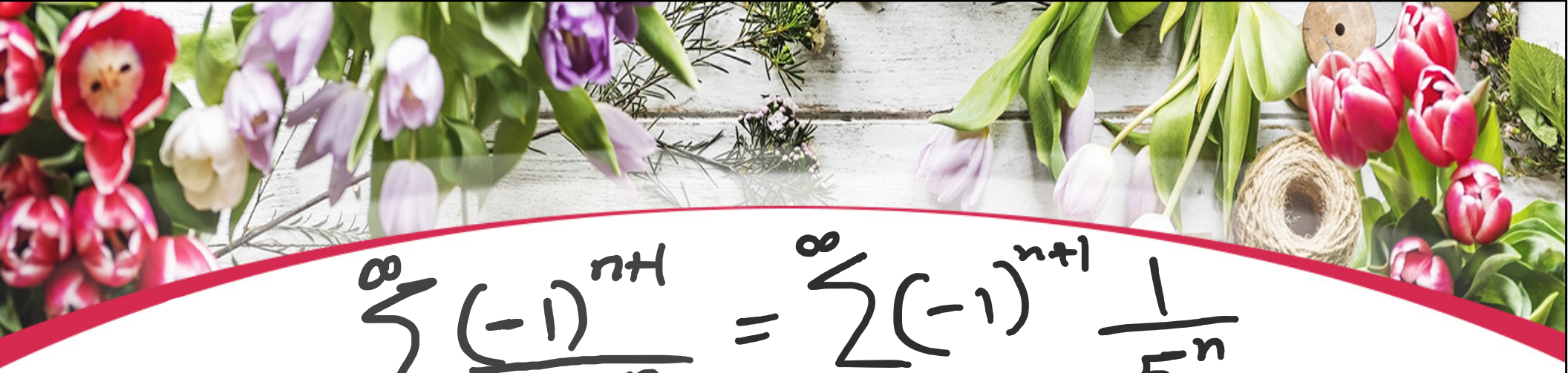
- The alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$$

satisfies i) $b_{n+1} < b_n$ because $\dfrac{1}{n+1} < \dfrac{1}{n}$

(ii) $\lim_{n \to \infty} b_n = \lim_{n \to \infty} \dfrac{1}{n} = 0$

- so the series is convergent by the Alternating Series Test.



Exercise: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{5^n} = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{5^n}$